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Purifying Interaction Effects with the Functional ANOVA: Reproducibility Study and Sensitivity Analysis

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Technical Report

Research Seminar in Explainable AI:
Reproducibility and Critical Evaluation of Methods

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Abstract

Additive models with pairwise interactions are widely used for interpretable machine learning, yet their component functions suffer from a fundamental identifiability problem: variance can be freely shifted between main effects and interactions without changing predictions, yielding contradictory explanations for the same model. Lengerich et al. (2020) proposed a purification algorithm based on the functional analysis of variance that enforces integrate-to-zero constraints via iterative mass-moving, producing a unique, canonical decomposition. This report reproduces their method for piecewise-constant models (depth-2 XGBoost ensembles) and extends the original work with two new experiments. First, we apply purification to the German Credit dataset under three weighting distributions (empirical, uniform, and independence-assumed), confirming that predictions remain invariant (RMSE = 2043.3 across all modes) while the attribution of variance to main effects versus interactions changes with the choice of weights. Second, we conduct a synthetic stress-test with controlled feature dependence ($\rho \in \{0, 0.3, 0.6, 0.9, 0.97\}$) and two ground-truth functions, measuring weighted L_2 distance, sign flip rate, and interaction variance share against the empirical-weight baseline. At $\rho = 0.97$, misspecified weights produce interaction surfaces with L_2 distances up to 0.90, sign flip rates of 17–34%, and variance attribution shifts from near 0% (empirical) to 73% (independence-assumed). These results demonstrate that while purification successfully resolves the identifiability problem, the resulting explanations are highly sensitive to the assumed weighting distribution when features are dependent, underscoring the need for careful weight specification and sensitivity analysis in practice.

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List of Acronyms

EBM explainable boosting machine.

fANOVA functional analysis of variance.

GA^{2}M generalized additive model with pairwise interactions.

GAM generalized additive model.

RMSE root mean squared error.

SHAP SHapley Additive exPlanations.

1. Introduction

1.1. Motivation

Interpretable machine learning models are increasingly deployed in high-stakes domains such as credit scoring, criminal justice, and healthcare, where understanding *why* a model makes a particular prediction is as important as the prediction itself (Molnar, 2022). Generalized additive models (GAMs) with pairwise interactions—sometimes called generalized additive models with pairwise interactions (GA^{2}Ms)—offer an attractive middle ground between black-box models and simple linear models (Lou, Caruana, Gehrke, and Hooker, 2013), decomposing predictions into univariate main effects $f_i(x_i)$ and bivariate interaction surfaces $f_{ij}(x_i, x_j)$.

However, this decomposition is **not unique**. One can add an arbitrary function $h(x_i)$ to the main effect f_i and subtract it from the interaction f_{ij} without changing predictions. This *identifiability problem* means the same model can yield contradictory interpretations (Lengerich et al., 2020).

1.2. Problem

The identifiability problem has practical consequences. In credit scoring, whether a feature's influence is classified as a "main effect" or an "interaction" affects regulatory explanations. In algorithmic fairness, interaction effects between sensitive attributes and other features can change the narrative around model bias (Angwin et al., 2016).

Lengerich et al. (2020) proposed *purification* as a solution: enforce the functional analysis of variance (fANOVA) integrate-to-zero constraints via an iterative mass-moving algorithm, yielding a canonical decomposition. However, the purified decomposition depends on a weighting distribution $w(x)$, and the original paper leaves largely unexplored how sensitive the purified effects are to this choice—particularly when features are dependent.

1.3. Approach and Research Questions

This report reproduces the purification algorithm for piecewise-constant models (depth-2 XGBoost ensembles) and extends it with two experiments probing weight sensitivity:

RQ1 Does the choice of weighting distribution ($\hat{w}_{\text{emp}}$, $\hat{w}_{\text{unif}}$, w_{indep}) qualitatively change the purified main effects and interactions on a real-world dataset (German Credit)?

RQ2 How does the divergence between purified effects under different weight specifications scale with the strength of feature dependence?

RQ3 At what level of dependence do misspecified weights produce qualitatively misleading interpretations (sign flips, variance attribution reversals)?

1.4. Outline

Chapter 2 introduces additive models, the fANOVA decomposition, and the integrate-to-zero constraints. Chapter 3 reviews related work. Chapter 4 describes the purification algorithm and our XGBoost decomposition. Chapters 5 and 6 detail the experimental setup and results. Chapter 7 interprets the findings, discusses ethical implications, and concludes. The source code is publicly available.[1]

[1] `https://github.com/HaykTarkhanyan/fanova_purification`

2. Background

2.1. Additive Models with Interactions

We consider models of the form

$$F(\mathbf{x}) = f_0 + \sum_{i=1}^{d} f_i(x_i) + \sum_{i<j} f_{ij}(x_i, x_j) + \cdots, \tag{2.1}$$

where $f_0 \in \mathbb{R}$ is an intercept, $f_i \colon \mathbb{R} \to \mathbb{R}$ are univariate *main effects*, and $f_{ij} \colon \mathbb{R}^2 \to \mathbb{R}$ are bivariate *interaction effects*. This class includes GAMs (Hastie and Tibshirani, 1986), GA^{2}Ms (Lou, Caruana, Gehrke, and Hooker, 2013), and explainable boosting machines (EBMs) (Nori et al., 2019). The appeal is interpretability: each f_i can be plotted as a curve and each f_{ij} as a surface.

2.2. The Identifiability Problem

Despite this appeal, the decomposition in Eq. (2.1) is **not unique**. For any function $h(x_i)$, one can replace

$$f_i(x_i) \leftarrow f_i(x_i) + h(x_i), \qquad f_{ij}(x_i, x_j) \leftarrow f_{ij}(x_i, x_j) - h(x_i),$$

without changing the overall predictor F. Without additional constraints, the attribution of variance to "main effects" versus "interactions" is arbitrary.

As a concrete example, consider $F(x_1, x_2) = x_1 \cdot x_2$ on $\{0, 1\}^2$. One parameterization places all variance in the interaction:

$$f_0 = 0, \quad f_1 = f_2 = 0, \quad f_{12}(x_1, x_2) = x_1 \cdot x_2.$$

Another (the fANOVA under uniform weights) shifts mass into the mains:

$$f_0 = \tfrac{1}{4}, \quad f_1(x_1) = \tfrac{1}{2}(x_1 - \tfrac{1}{2}), \quad f_2(x_2) = \tfrac{1}{2}(x_2 - \tfrac{1}{2}), \quad f_{12}(x_1, x_2) = (x_1 - \tfrac{1}{2})(x_2 - \tfrac{1}{2}).$$

Both represent the same function, yet the first attributes all variance to the interaction while the second distributes it equally among f_1, f_2, and f_{12}.

2.3. Functional ANOVA Decomposition

The fANOVA provides a canonical resolution. Given a weight function $w(\mathbf{x}) \geq 0$ with $\int w(\mathbf{x})\, d\mathbf{x} = 1$, it decomposes F into **orthogonal** components:

$$F(\mathbf{x}) = f_0 + \sum_{u \subseteq \{1,\ldots,d\}} f_u(\mathbf{x}_u), \tag{2.2}$$

subject to the *integrate-to-zero* constraint:

$$\boxed{\mathbb{E}_w[f_u(\mathbf{x}_u) \mid \mathbf{x}_{u \setminus \{i\}}] = 0 \quad \forall\, i \in u.} \tag{2.3}$$

For a main effect, this requires $\mathbb{E}_w[f_i] = 0$; for an interaction, $\mathbb{E}_w[f_{ij} \mid x_i] = 0$ and $\mathbb{E}_w[f_{ij} \mid x_j] = 0$. The resulting decomposition is unique and yields orthogonality ($\mathbb{E}_w[f_u \cdot f_v] = 0$ for $u \neq v$) and variance decomposition ($\mathrm{Var}_w(F) = \sum_u \mathrm{Var}_w(f_u)$) (Hooker, 2007; Sobol', 2001).

2.4. Piecewise-Constant Case: Tensors and Slice Mean-Zero

For tree ensembles, each feature x_i is discretized into B_i bins. Main effects become vectors $\mathbf{f}_i \in \mathbb{R}^{B_i}$ and interactions become matrices $\mathbf{F}_{ij} \in \mathbb{R}^{B_i \times B_j}$. The integrate-to-zero constraint becomes the *weighted slice mean-zero* constraint:

$$\text{Row constraint:} \quad \sum_{k=1}^{B_j} \mathbf{F}_{ij}[r,k] \cdot w_j[k] = 0 \quad \forall\, r, \tag{2.4}$$

$$\text{Column constraint:} \quad \sum_{k=1}^{B_i} \mathbf{F}_{ij}[k,c] \cdot w_i[k] = 0 \quad \forall\, c. \tag{2.5}$$

2.5. Weight Estimation

The fANOVA depends on the weighting distribution w, which must be estimated in practice. Lengerich et al. (2020) discuss several estimators:

Uniform ($\hat{w}_{\text{unif}}$): Equal weight to all bin combinations: $w(\mathbf{x}) \propto 1$.

Empirical ($\hat{w}_{\text{emp}}$): Weights proportional to observed bin counts: $w(\mathbf{x}) \propto \hat{p}(\mathbf{x})$.

Laplace ($\hat{w}_{\text{lap}}$): Smoothed blend $\hat{w}_{\text{lap}} \propto \hat{w}_{\text{emp}} + \hat{w}_{\text{unif}}$, adding pseudo-counts to avoid zero-weight bins.

Independence-assumed (w_{indep}): Product of marginals: $w_{\text{indep}}(\mathbf{x}) = \prod_i \hat{p}(x_i)$.

Permutation joint (w_{perm}): Empirical joint on data with columns independently shuffled.

The choice has no effect on predictions but profoundly affects variance attribution, especially when features are dependent. Figure 2.1 contrasts uniform and empirical marginal weights for a single feature.

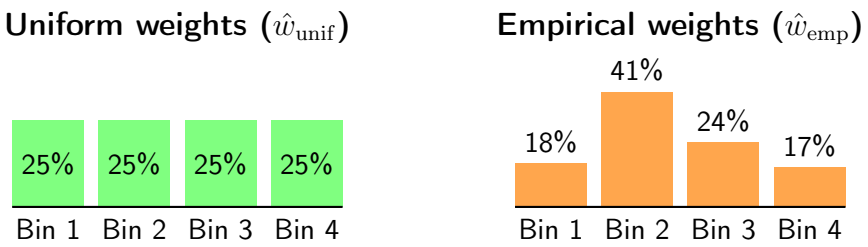


Figure 2.1.: Marginal weights for a single feature. Under $\hat{w}_{\text{unif}}$ (left), all bins are weighted equally. Under $\hat{w}_{\text{emp}}$ (right), weights reflect the data distribution.

2.5.1. Joint Weights and Feature Dependence

For pairwise interactions, the joint weight matrix $\mathbf{W}_{ij}$ determines the slice mean-zero constraint. Figure 2.2 compares $\hat{w}_{\text{emp}}$ with w_{indep} for two correlated features: under $\hat{w}_{\text{emp}}$, mass concentrates along the diagonal; under w_{indep}, it spreads as the product of marginals. Since purification subtracts weighted slice means, these produce different purified effects.

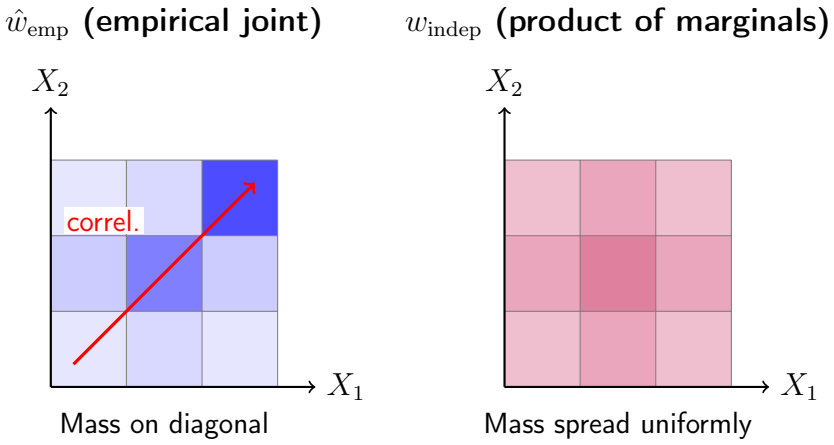


Figure 2.2.: Joint weight matrices for two correlated features. Under $\hat{w}_{\text{emp}}$ (left), mass concentrates on the diagonal. Under w_{indep} (right), mass is the product of marginals.

3. Related Work

3.1. Generalized Additive Models and Interpretability

GAMs, introduced by Hastie and Tibshirani (1986), model the response as a sum of univariate functions. Lou, Caruana, and Gehrke (2012) and Lou, Caruana, Gehrke, and Hooker (2013) proposed GA^{2}Ms, augmenting GAMs with pairwise interaction terms, and Microsoft's EBM (Nori et al., 2019) brought these to production use. More recently, Agarwal et al. (2021) introduced Neural Additive Models (NAMs), which inherit the additive structure (and thus the identifiability problem). Chang et al. (2021) showed that different GAM training algorithms produce substantially different shape functions for the same data—directly analogous to our finding that different weight specifications yield different purified effects.

3.2. Functional ANOVA and Sensitivity Analysis

The fANOVA decomposition has roots in global sensitivity analysis. Sobol' (2001) introduced Sobol' indices based on the orthogonal decomposition of a function under a product measure (assuming independent inputs). Hooker (2007) extended the framework to dependent variables, showing that the stan-

dard decomposition can misattribute interaction variance to main effects and vice versa when features are correlated—the phenomenon our experiments quantify.

3.3. Feature Interactions and Their Detection

Friedman and Popescu (2008) proposed the H-statistic for measuring interaction strength, conceptually related to the interaction variance share we compute after purification. Lundberg and Lee (2017) introduced SHapley Additive exPlanations (SHAP) interaction values, which decompose pairwise effects at the individual prediction level rather than as global shape functions, making them complementary to purification. Hooker et al. (2021) and Strobl et al. (2008) showed that permutation-based importance measures are unreliable when features are dependent—paralleling our finding that the weighting distribution w plays an analogous role to the reference distribution.

3.4. Partial Dependence and Accumulated Local Effects

Friedman (2001) introduced partial dependence plots (PDPs), which implicitly assume feature independence and can mislead when features are correlated. Apley and Zhu (2020) developed accumulated local effects (ALE) plots as an alternative that avoids extrapolation by restricting attention to observed feature combinations. Both face the same marginal-vs-conditional tension that our work addresses through weight specification.

3.5. Purification and Identifiability

Lengerich et al. (2020) identified the non-identifiability of additive-with-interaction models and proposed the mass-moving purification algorithm. They proved convergence guarantees and demonstrated the algorithm on COMPAS, pneumonia risk, and recidivism datasets, showing that purification qualitatively changes which features appear important and how interactions are attributed.

However, their analysis focused on comparing purified versus unpurified effects, rather than studying sensitivity to the weighting distribution w. This gap—understanding how much purified effects change when w is varied—is the primary contribution of our work.

3.6. Algorithmic Fairness and the German Credit Dataset

The German Credit dataset (Dua and Graff, 2019) is a standard benchmark in fair lending research, containing sensitive attributes such as age and sex. The COMPAS recidivism system (Angwin et al., 2016) highlighted how model interpretations can have profound societal consequences, with debates centering on how "bias" is measured and attributed. The purification algorithm is directly relevant: by choosing different weighting distributions, a practitioner could shift variance between main effects involving sensitive attributes and their interactions, raising questions about the trustworthiness of purification-based audits (Section 7.3).

4. Methods

This chapter describes the mass-moving purification algorithm and our implementation for XGBoost depth-2 ensembles.

4.1. Mass-Moving Purification Algorithm

The purification algorithm enforces the weighted slice mean-zero constraints (Eqs. (2.4) and (2.5)) by iteratively "moving mass" from higher-order tensors to lower-order tensors.

4.1.1. Purify-Matrix: A Single Interaction Tensor

For a pairwise interaction matrix $\mathbf{F}_{ij} \in \mathbb{R}^{B_i \times B_j}$ with marginal weight vectors $\mathbf{w}_i \in \mathbb{R}^{B_i}$ and $\mathbf{w}_j \in \mathbb{R}^{B_j}$, each sweep proceeds as follows:

1. **Row sweep** (enforcing the row constraint): for each row $r = 1, \ldots, B_i$:
   a) Compute the weighted mean across columns: $\bar{m}_r = \frac{\sum_{k=1}^{B_j} \mathbf{F}_{ij}[r,k] \cdot w_j[k]}{\sum_{k=1}^{B_j} w_j[k]}$.
   b) Subtract from the interaction row: $\mathbf{F}_{ij}[r,k] \leftarrow \mathbf{F}_{ij}[r,k] - \bar{m}_r$ for all k.
   c) Add to the main effect of X_i: $f_i[r] \leftarrow f_i[r] + \bar{m}_r$.
2. **Column sweep** (enforcing the column constraint): for each column $c = 1, \ldots, B_j$:

a) Compute the weighted mean across rows: $\bar{m}_c = \frac{\sum_{k=1}^{B_i} \mathbf{F}_{ij}[k,c] \cdot w_i[k]}{\sum_{k=1}^{B_i} w_i[k]}$.

b) Subtract from the interaction column: $\mathbf{F}_{ij}[k,c] \leftarrow \mathbf{F}_{ij}[k,c] - \bar{m}_c$ for all k.

c) Add to the main effect of X_j: $f_j[c] \leftarrow f_j[c] + \bar{m}_c$.

3. Repeat sweeps until $\max_{r,c}(|\bar{m}_r|, |\bar{m}_c|) \leq \varepsilon$.

This is a Gauss–Seidel-style iteration: updating rows changes the column means, so the process iterates. The key invariant is that $F(\mathbf{x})$ is unchanged after each step, since the subtracted mass is added to a lower-order term. If $\sum_k w_j[k] = 0$ for a particular slice, the weighted mean is undefined; we treat this as $\bar{m} = 0$ (no mass is moved).

4.1.2. Worked Example

Figure 4.1 illustrates one row sweep on a 3×3 interaction matrix with uniform weights.

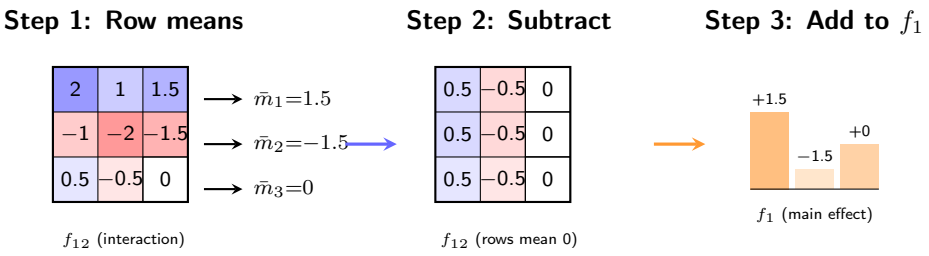


Figure 4.1.: One row sweep of the mass-moving algorithm on a 3×3 interaction matrix (uniform weights). Step 1: compute row means. Step 2: subtract from interaction. Step 3: add to f_1. The overall function $f_1(x_1) + f_{12}(x_1, x_2)$ is unchanged.

4.1.3. Purify-All: Processing All Tensors

To purify a complete additive model, we process tensors in descending order of interaction order. For depth-2 models: purify all pairwise interaction matrices first, then center all main effects (subtracting their weighted mean and adding it to the intercept).

Algorithm 1 Mass-moving purification (Purify-All)

Require: Tensors $\{T_u\}$ for all subsets u, weights $\{w_u\}$, tolerance ε
Ensure: Purified tensors satisfying slice mean-zero constraints

```
Sort keys u by |u| descending
for each u in sorted order do
    repeat
        max_residual ← 0
        for each axis a corresponding to variable i ∈ u do
            for each slice along the other axes do
                m̄ ← weighted mean of slice using w_i
                T_u[slice] ← T_u[slice] − m̄
                T_{u\{i}}[corresponding entry] ← T_{u\{i}}[...] + m̄
                max_residual ← max(max_residual, |m̄|)
            end for
        end for
    until max_residual ≤ ε
end for
```

4.1.4. Convergence Properties

Lengerich et al. (2020) proved that for uniform or factored weight distributions, purification converges in a single pass (row sweeps do not alter column means). For arbitrary positive weights, convergence to tolerance ε requires $O(\log(M_0) - \log(\varepsilon))$ iterations. Figure 4.2 confirms this empirically: uniform weights converge in one sweep; random weights reach 10^{-10} within 5–10 sweeps.

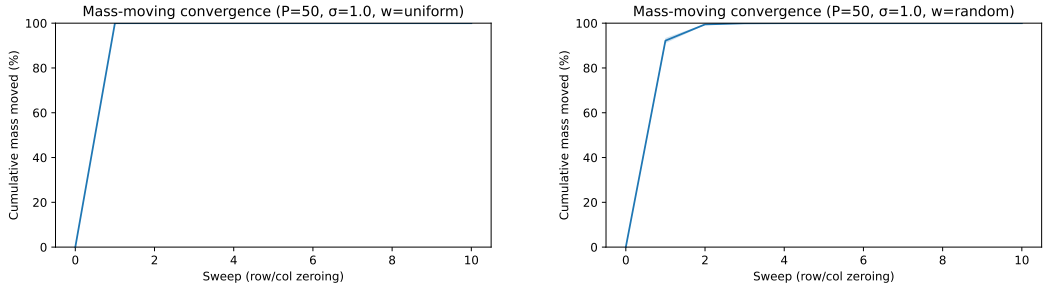


Figure 4.2.: Convergence on random 50×50 matrices ($\sigma = 1.0$). Left: uniform weights—one sweep. Right: random weights—exponential decay over 5–10 sweeps.

4.2. XGBoost Depth-2 Decomposition

For an XGBoost model restricted to depth ≤ 2, each root-to-leaf path uses at most two features, enabling exact decomposition into main effects and pairwise interactions.

4.2.1. Binning

The bin boundaries for each feature x_j are defined by the union of all split thresholds across all trees, partitioning the axis into $K + 1$ bins. Each input is mapped to a bin index via searchsorted, matching XGBoost's "left if $x < t$" convention.

4.2.2. Tensor Construction

The ensemble is decomposed into:

- An intercept $T_{()} = \text{base_score}$.
- Main effect vectors $T_{(i)} \in \mathbb{R}^{B_i}$, from leaf paths using only feature i.

- Interaction matrices $T_{(i,j)} \in \mathbb{R}^{B_i \times B_j}$, from leaf paths using features i and j.

For each tree, we extract all root-to-leaf paths; each path defines a rectangular region in bin space, and the leaf value is added to all tensor entries within that region. A depth-2 tree can use up to three features across branches, but each path uses at most two, so the tree decomposes into pairwise interactions and main effects at the path level.

4.2.3. Exactness

This decomposition is exact:

$$F(\mathbf{x}) = T_{()} + \sum_i T_{(i)}[\text{bin}_i(\mathbf{x})] + \sum_{i<j} T_{(i,j)}[\text{bin}_i(\mathbf{x}), \text{bin}_j(\mathbf{x})]$$

reproduces XGBoost's predictions exactly, verified in every experiment.

4.3. Prediction Invariance

Purification preserves predictions by construction: subtracting $\bar{m}$ from a row of $\mathbf{F}_{ij}$ and adding it to f_i leaves

$$f_i[\text{bin}_i] + \mathbf{F}_{ij}[\text{bin}_i, \text{bin}_j] = (f_i[\text{bin}_i] + \bar{m}) + (\mathbf{F}_{ij}[\text{bin}_i, \text{bin}_j] - \bar{m}) = \text{unchanged}.$$

We verify this empirically by checking $\max_{\mathbf{x}} |F_{\text{model}}(\mathbf{x}) - F_{\text{purified}}(\mathbf{x})| < 10^{-2}$.

5. Experiments

We conduct two experiments to investigate the sensitivity of purified effects to the weighting distribution. Both share the same pipeline: fit a depth-2 XGBoost (Chen and Guestrin, 2016) model, decompose it into tensors, and purify under multiple weight specifications.

5.1. Experimental Setup

All experiments are implemented in Python 3.10+ using NumPy, pandas, scikit-learn, XGBoost, and Matplotlib. The purification algorithm and XGBoost decomposition are implemented from scratch as the `purify_fanova` library (source code at `https://github.com/HaykTarkhanyan/fanova_purification`).

We use purification tolerance $\varepsilon = 10^{-10}$ and XGBoost with `max_depth=2`, `learning_rate=0.1`, no L2 regularization, and full subsampling. The depth-2 constraint ensures exact decomposition into main effects and pairwise interactions (Section 4.2). Prediction invariance is verified in every run.

5.2. Experiment 1: German Credit Dataset

5.2.1. Data

We use the German Credit dataset (Dua and Graff, 2019): 1000 samples, 20 features describing loan applicants. We predict `Credit amount` as a regression target. Categorical features are one-hot encoded (yielding 28 inputs), with a 75%/25% train/test split.

5.2.2. Model

We train an XGBoost regressor with 120 trees, `max_depth=2`, and `learning_rate=0.1` (full hyperparameters in Table A.1).

5.2.3. Purification Modes

The same trained model is purified under three weighting distributions:

- $\hat{w}_{\text{emp}}$: empirical joint distribution from training data.
- $\hat{w}_{\text{unif}}$: uniform weights over all bin combinations.
- w_{indep}: product of empirical marginals ($w_{\text{indep}}(b_i, b_j) = \hat{p}(b_i) \cdot \hat{p}(b_j)$).

5.2.4. Evaluation

We report test root mean squared error (RMSE) (identical across modes, confirming prediction invariance), top effects ranked by variance, and visual comparison of main effect curves and interaction heatmaps.

5.3. Experiment 2: Synthetic Stress-Test Under Controlled Dependence

5.3.1. Motivation

The German Credit experiment demonstrates weight sensitivity on real data but does not allow systematic control of feature dependence. The stress-test generates bivariate data with known dependence and ground-truth interactions.

5.3.2. Data Generation

We generate bivariate data with controlled linear-Gaussian dependence:

$$X_1 \sim \mathcal{N}(0,1), \qquad \varepsilon \sim \mathcal{N}(0,1) \text{ independent}, \tag{5.1}$$

$$X_2 = \rho \cdot X_1 + \sqrt{1-\rho^2} \cdot \varepsilon, \tag{5.2}$$

where $\rho \in \{0.0, 0.3, 0.6, 0.9, 0.97\}$. By construction, $\mathrm{Corr}(X_1, X_2) = \rho$ with unit marginal variances. We generate $n = 4000$ samples per setting.

5.3.3. Ground-Truth Functions

F1 (Multiplicative): $F(X_1, X_2) = X_1 + X_2 + 1.5 \cdot X_1 X_2 + \varepsilon_{\text{noise}}$. Smooth "saddle" interaction with well-behaved marginal structure.

F2 (XOR-like): $F(X_1, X_2) = 2 \cdot \mathrm{sign}(X_1) \cdot \mathrm{sign}(X_2) + 0.5X_1 + 0.5X_2 + \varepsilon_{\text{noise}}$. Sharp checkerboard interaction ($+2$ in quadrants I/III, -2 in II/IV), expected to be more sensitive to weight misspecification.

Both include Gaussian noise with $\sigma = 0.2$.

5.3.4. Model and Purification

For each (ρ, function, random seed) combination, we fit a depth-2 XGBoost regressor with 60 trees and purify under four weight modes: $\hat{w}_{\text{emp}}$, $\hat{w}_{\text{unif}}$, w_{indep}, and w_{perm}. We use 3 random seeds per setting and report means.

5.3.5. Metrics

All metrics measure divergence from the $\hat{w}_{\text{emp}}$-purified baseline.

Weighted L_2 distance: Overall surface difference:

$$d_{L_2} = \sqrt{\sum_{i,j} w_{ij}^{\text{emp}} \left(f_{ij}^{\text{emp}} - f_{ij}^{\text{other}}\right)^2}.$$

Sign flip rate: Fraction of bins whose sign differs between weight modes (excluding near-zero bins with $|f| < 10^{-12}$). A sign flip is a qualitative reversal in interpretation.

Interaction variance share: $\text{IVS} = \text{Var}_w(f_{12}) / \text{Var}_w(F)$, computed with each mode's own weights. Determines the narrative: "Is this model dominated by main effects or interactions?"

6. Results

6.1. Experiment 1: German Credit

6.1.1. Prediction Invariance

The test RMSE is 2043.3 across all three weighting modes ($\hat{w}_{\text{emp}}$, $\hat{w}_{\text{unif}}$, w_{indep}), confirming that purification does not alter predictions. The maximum absolute prediction error between the original XGBoost model and the purified tensor representation is below 10^{-2} in all cases.

6.1.2. Dominant Effects

Tables 6.1 and 6.2 show the top main effects and interactions ranked by variance under $\hat{w}_{\text{emp}}$. `Duration` dominates (variance $\approx 2.9 \times 10^6$); the strongest interaction is `Duration` $\times$ `Purpose_vacation` ($\approx$ 29,522).

Table 6.1.: Top main effects ranked by weighted variance under $\hat{w}_{\text{emp}}$.

Feature	Variance
Duration	2,906,619
Job_3	247,365
Purpose_vacation	93,542
Age	79,656
Purpose_radio/TV	43,529
Purpose_education	9,319

Table 6.2.: Top pairwise interactions ranked by weighted variance under $\hat{w}_{\text{emp}}$.

Feature pair	Variance
Duration × Purpose_vacation	29,522
Age × Duration	17,139
Duration × Job_3	7,498
Saving_moderate × Purpose_education	5,888
Sex_female × Purpose_education	3,015

6.1.3. Main Effect Sensitivity to Weights

Figure 6.1 shows the purified main effect curves under the three weight modes. The overall shape is preserved, but magnitudes and vertical offsets differ, reflecting variance reallocation between mains and interactions.

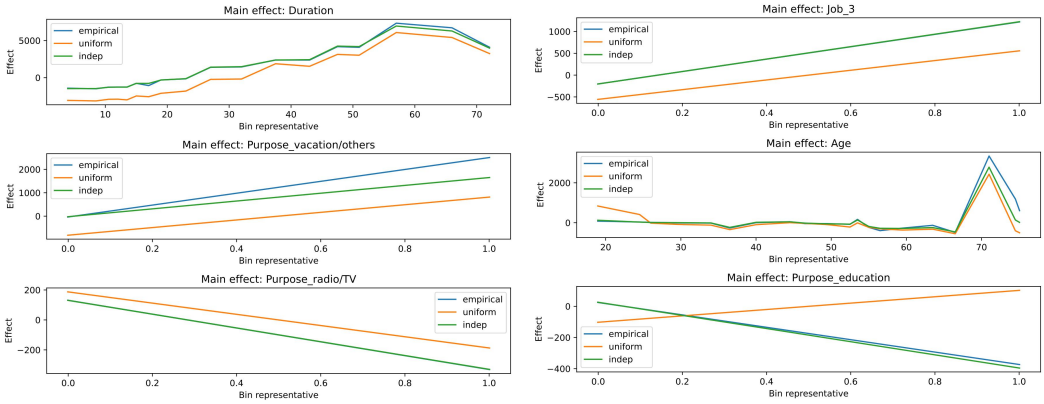


Figure 6.1.: Purified main effect curves for the top features under $\hat{w}_{\text{emp}}$ (blue), $\hat{w}_{\text{unif}}$ (orange), and w_{indep} (green). Predictions are identical across all modes.

6.1.4. Interaction Heatmaps

Figure 6.2 shows the top interaction surface before and after purification under each weight mode. The unpurified interaction contains substantial marginal structure; after purification, the qualitative pattern of the residual differs across weight modes.

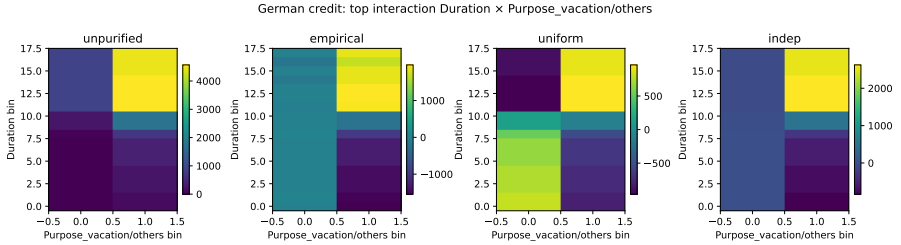


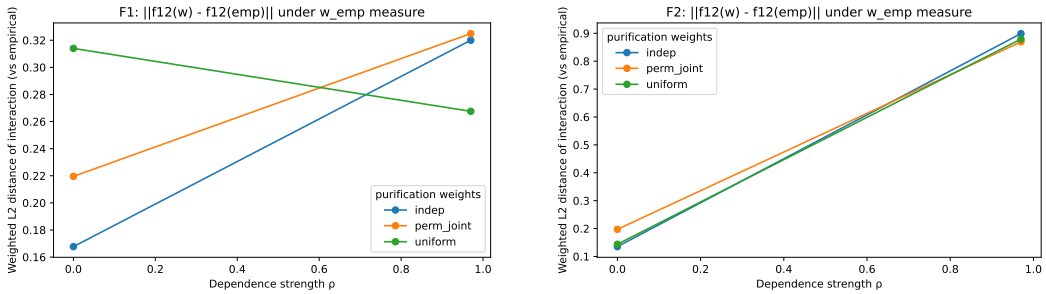
Figure 6.2.: Top interaction surface (Duration $\times$ Purpose_vacation) before purification (leftmost) and after purification under $\hat{w}_{\text{emp}}$, $\hat{w}_{\text{unif}}$, and w_{indep}.

6.2. Experiment 2: Stress-Test Under Controlled Dependence

All metrics are reported relative to the $\hat{w}_{\text{emp}}$-purified baseline. For each metric, we show F1 (multiplicative, expected to be more robust) and F2 (XOR-like, expected to be more sensitive).

6.2.1. L_2 Distance vs. Dependence Strength

For F1 (Fig. 6.3a), the L_2 distance remains moderate (0.27–0.32 at $\rho = 0.97$). For F2 (Fig. 6.3b), the picture changes dramatically: at $\rho = 0.97$, the L_2 distance reaches 0.87–0.90—the purified surface is nearly unrecognizable compared to the $\hat{w}_{\text{emp}}$ baseline.



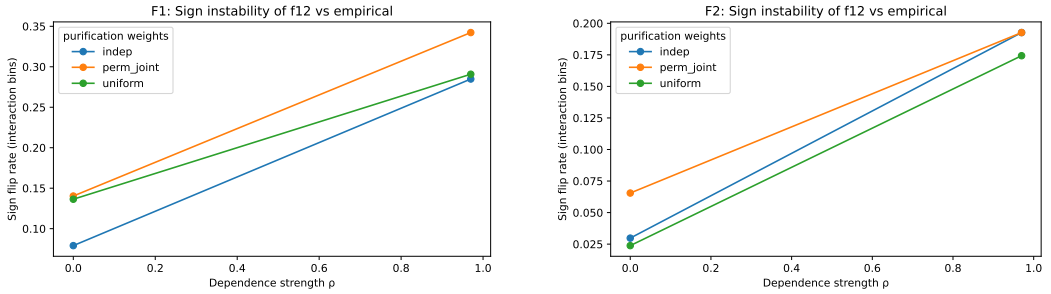
(a) F1 (multiplicative)

(b) F2 (XOR-like)

Figure 6.3.: Weighted L_2 distance from the $\hat{w}_{\text{emp}}$ baseline. F1 (left) remains moderate; F2 (right) approaches 0.9 at high ρ.

6.2.2. Sign Flip Rate

For F1 (Fig. 6.4a), sign flips are negligible at low ρ but reach 29–34% at $\rho = 0.97$—roughly a third of bins reverse their interpretation. For F2 (Fig. 6.4b), the rate is lower (17–19%) despite F2's larger L_2 distance, because the XOR-like interaction has large absolute values in all quadrants—sign flips occur mainly near the zero-crossing boundaries.



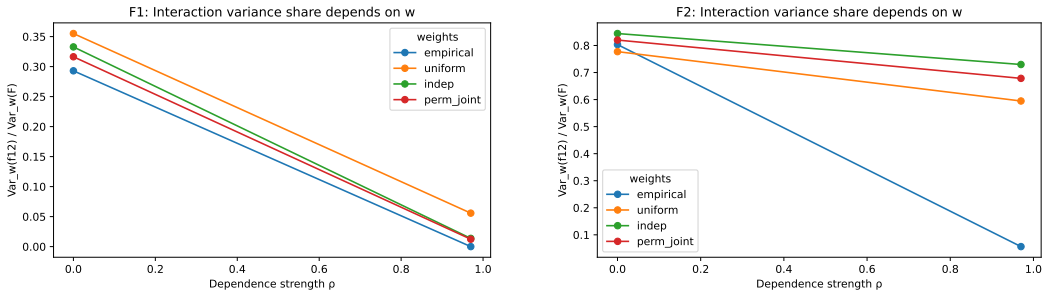
(a) F1 (multiplicative)

(b) F2 (XOR-like)

Figure 6.4.: Fraction of interaction bins whose sign differs from the $\hat{w}_{\text{emp}}$ baseline. F1 (left) reaches 29–34%; F2 (right) reaches 17–19% at high ρ.

6.2.3. Interaction Variance Share

The interaction variance share determines the high-level narrative: "Is this model dominated by main effects, or does the interaction matter?" For F1 (Fig. 6.5a), the share under $\hat{w}_{\text{emp}}$ drops from 29% at $\rho = 0$ to essentially 0% at $\rho = 0.97$, while $\hat{w}_{\text{unif}}$ and w_{indep} retain 1–6%. For F2 (Fig. 6.5b), the divergence is dramatic: at $\rho = 0.97$, $\hat{w}_{\text{emp}}$ reports 5.6% while w_{indep} reports **73%**—the same model and data yield diametrically opposed narratives depending solely on the weight specification.



(a) F1 (multiplicative)

(b) F2 (XOR-like)

Figure 6.5.: Interaction variance share. F1 (left): $\hat{w}_{\text{emp}}$ drops to ≈0% at high ρ. F2 (right): $\hat{w}_{\text{emp}}$ reports 5.6% while w_{indep} reports 73%.

6.2.4. Main Effects at Strong Dependence

Where does the "missing" interaction variance go? Figure 6.6 shows the main effect curves for F1 at $\rho = 0.97$: under $\hat{w}_{\text{emp}}$, the mains are steeper (absorbing nearly all variance), while under w_{indep} and $\hat{w}_{\text{unif}}$ they are flatter.

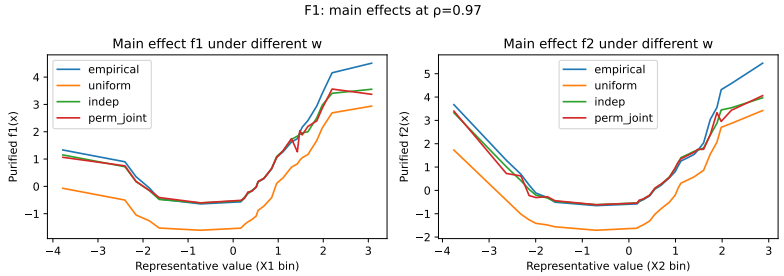


Figure 6.6.: F1 main effect curves at $\rho = 0.97$. $\hat{w}_{\text{emp}}$ curves are steeper; w_{indep} and $\hat{w}_{\text{unif}}$ retain more interaction structure.

6.2.5. Interaction Heatmaps at Strong Dependence

Figure 6.7 shows the F1 interaction surface at $\rho = 0.97$: under $\hat{w}_{\text{emp}}$ the residual is nearly flat, while w_{indep} and $\hat{w}_{\text{unif}}$ partially preserve the saddle structure.

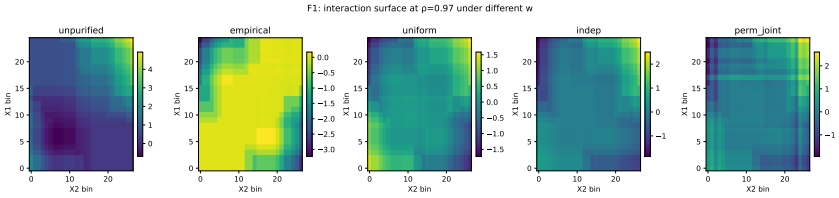


Figure 6.7.: F1 interaction surfaces at $\rho = 0.97$: unpurified (leftmost), then purified under each weight mode.

Figure 6.8 shows F2, where the contrast is most striking: the checkerboard pattern is clearly visible under w_{indep} and $\hat{w}_{\text{unif}}$ but largely absent under $\hat{w}_{\text{emp}}$.

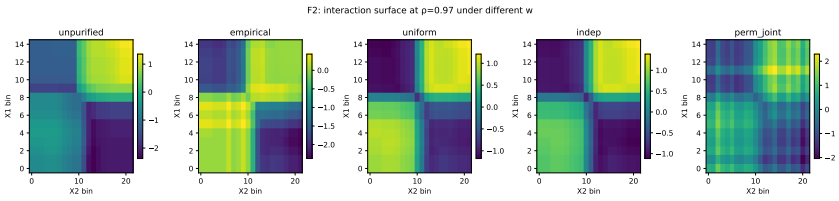


Figure 6.8.: F2 interaction surfaces at $\rho = 0.97$. w_{indep} shows a strong checkerboard; $\hat{w}_{\text{emp}}$ shows a much weaker pattern.

6.2.6. Summary of Stress-Test Results

Table 6.3 summarizes the key metrics at the extremes of dependence. At $\rho = 0$, all weight modes agree; at $\rho = 0.97$, they diverge substantially. F2 is more sensitive than F1 on every metric except sign flip rate, where F1's smoother surface leads to more sign reversals near zero crossings.

Table 6.3.: Summary of stress-test metrics at $\rho = 0$ and $\rho = 0.97$ for both ground-truth functions. Interaction variance share is given for each weight mode; L_2 distance and sign flip rate are reported for non-empirical modes against the $\hat{w}_{\text{emp}}$ baseline.

	F1 (multiplicative)		**F2 (XOR-like)**	
Metric / Mode	ρ=0	ρ=0.97	ρ=0	ρ=0.97
Interaction variance share (%)				
$\hat{w}_{\text{emp}}$	29	≈0	80	5.6
$\hat{w}_{\text{unif}}$	36	5.6	78	59
w_{indep}	33	1.4	84	73
L_2 distance (vs. $\hat{w}_{\text{emp}}$)				
uniform	0.31	0.27	0.14	0.88
indep	0.17	0.32	0.14	0.90
perm_joint	0.22	0.32	0.20	0.87
Sign flip rate (%, at $\rho = 0.97$)				
uniform	—	29	—	17
indep	—	28	—	19
perm_joint	—	34	—	19

7. Discussion and Conclusion

This chapter interprets the experimental results in the context of our three research questions, discusses broader implications, and outlines future directions.

7.1. Answering the Research Questions

7.1.1. RQ1: How Sensitive Are Purified Effects to the Weighting Distribution?

Sensitivity depends strongly on the nature of the interaction and the degree of feature dependence. In the German Credit experiment (Section 5.2), the three weight modes produce identical test RMSE (2043.3) but visibly different main effect curves and interaction heatmaps (Figs. 6.1 and 6.2). The synthetic stress-test (Section 5.3) provides sharper evidence: at $\rho = 0.97$, the L_2 distance reaches 0.90, the interaction variance share swings from 5.6% to 73%, and up to 34% of bins change sign (Table 6.3). We conclude that purified effects can be **extremely sensitive** to the weighting distribution when features are dependent and the interaction structure is non-multiplicative.

7.1.2. RQ2: Under What Conditions Does Sensitivity Arise?

Two conditions amplify weight sensitivity:

Feature dependence. When $\rho = 0$, all weight modes converge. Divergence increases monotonically with ρ across all metrics (Figs. 6.3a, 6.3b, 6.5a and 6.5b). This is expected: when features are independent, the integrate-to-zero constraint under $\hat{w}_{\text{emp}}$ coincides with w_{indep}; dependence breaks this coincidence.

Interaction structure. The XOR-like function F2 is far more sensitive than the multiplicative F1 across all metrics (Table 6.3). The multiplicative interaction has smooth marginal structure that absorbs mass gradually, while the XOR-like interaction's sharp non-linearity leads to large disagreements in what counts as "main effect" versus "interaction." For models with discrete or threshold-like interactions (common in tree ensembles on categorical features), weight sensitivity is likely most severe.

7.1.3. RQ3: What Are the Practical Implications?

1. **Always report the weight specification.** A purified effect plot without specifying w is incomplete.
2. **Compare multiple weight modes.** Agreement signals robustness; disagreement signals entanglement with feature dependence.
3. **Treat large variance share differences as a warning.** An unstable interaction story should not be presented as definitive.
4. **Be cautious with dependent features.** Any single purification result should be interpreted with care when features are strongly correlated.

7.2. Interpretation of Weight Modes

Each weight mode encodes a different answer to "what does it mean for an interaction to have zero marginal effect?" Under $\hat{w}_{\text{emp}}$, the constraint is evaluated against the observed data—natural for asking whether an interaction

contributes beyond its marginals in the data we observe, but sensitive to dependence artifacts. Under w_{indep} (and w_{perm}), the constraint assumes independent features—natural for a causal "what if we varied each feature independently?" perspective, but requires evaluating the model on out-of-distribution inputs. Under $\hat{w}_{\text{unif}}$, all bin combinations are treated equally, ignoring the data distribution entirely. None is universally correct; the choice depends on the scientific question, which is precisely the problem: the fANOVA does not resolve the interpretive ambiguity—it merely makes it explicit through w.

7.3. Potential for Misuse and Ethical Concerns

The weight sensitivity we have documented creates an opportunity for *explanation shopping*—selectively choosing the weight specification that produces the most favorable narrative.

7.3.1. Manipulating Variance Attribution

The interaction variance share can range from 5.6% to 73% for the same model (Table 6.3). A practitioner wanting to downplay an interaction (e.g., between a sensitive attribute and another feature) can choose a weight mode that absorbs interaction variance into the mains, and vice versa. In algorithmic fairness, this is particularly concerning: under $\hat{w}_{\text{emp}}$, a Sex×Duration interaction might appear negligible, while under w_{indep} it might be substantial—neither conclusion is "wrong," but reporting only one is misleading.

7.3.2. The German Credit Example

The interaction `Sex_female` × `Purpose_education` has empirical-weight variance $\approx$ 3,015 (Table 6.2). Under $\hat{w}_{\text{emp}}$, the weights reflect the observed data (most applicants are male, few apply for education loans), causing marginal

structure to be absorbed into the mains—resulting story: *"the model treats sex similarly regardless of loan purpose."* Under w_{indep}, the $(1, 1)$ cell is re-weighted to $\hat{p}(\text{female}) \cdot \hat{p}(\text{education})$, potentially leaving a larger residual interaction—resulting story: *"the model treats female education-loan applicants differently from what the main effects predict."* A regulator would reach very different conclusions about sex-based discrimination depending on which result they are shown.

7.3.3. Broader Implications

The ability to produce different explanations from the same model undermines the trust that interpretability methods are designed to build, paralleling concerns raised by Chang et al. (2021) for GAM training algorithms. Our contribution is to show that even *after* purification, a degree of freedom remains.

We do not argue that purification should be abandoned—it provides a principled framework that resolves the worst forms of non-identifiability. Rather, we argue that:

1. The weight specification should be treated as a **modeling choice** that must be justified and reported.
2. Regulatory frameworks should require disclosure of the weight specification used for any purification-based explanation.
3. Researchers should present results under multiple weight modes to demonstrate robustness.

7.4. Limitations

- **Depth-2 models only.** Deeper trees with higher-order interactions may exhibit different sensitivity patterns.

- **Two-variable stress-test.** Higher dimensions may introduce additional phenomena (e.g., confounding through a third variable).
- **Single real-world dataset.** Larger datasets with stronger dependence structures might amplify or dampen the observed effects.
- **Linear-Gaussian dependence.** Non-linear dependence structures (e.g., copulas with tail dependence) might produce different sensitivity patterns.
- **No ground-truth for the real-world experiment.** The stress-test addresses this partially, but only for synthetic data.

7.5. Conclusion

This report has investigated the sensitivity of the functional ANOVA purification algorithm to the weighting distribution used to enforce the integrate-to-zero constraints. All differences arise while predictions remain perfectly invariant. This is a fundamental limitation of the fANOVA framework when features are dependent, not a bug in any implementation.

7.6. Outlook

Higher-order interactions. Extending to higher-order interactions (depth > 2 trees) would require tensor decompositions of order three or more, where weight sensitivity may be even more severe.

Alternative weight estimation. Kernel density estimators, copula-based weights, or learned density models could provide better approximations to the true data-generating distribution.

Robustness certificates. Reporting the range of purified effects across a family of plausible weight distributions would provide robustness certificates for interpretations.

Beyond tree ensembles. The purification framework applies to any additive model with interactions, including neural additive models (Agarwal et al., 2021) and spline-based GA2Ms.

Regulatory standards. Model documentation standards should include the weight specification used for any purification-based explanation, with guidelines for when weight sensitivity is likely to be a concern.

8. Bibliography

Agarwal, Rishabh, Levi Melnick, Nicholas Frosst, Xuezhou Zhang, Ben Lengerich, Rich Caruana, and Geoffrey E. Hinton (2021). "Neural Additive Models: Interpretable Machine Learning with Neural Nets". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 34.

Angwin, Julia, Jeff Larson, Surya Mattu, and Lauren Kirchner (2016). "Machine Bias". In: *ProPublica*. `https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing`.

Apley, Daniel W. and Jingyu Zhu (2020). "Visualizing the Effects of Predictor Variables in Black Box Supervised Learning Models". In: *Journal of the Royal Statistical Society: Series B* 82.4, pp. 1059–1086.

Chang, Chun-Hao, Sarah Tan, Ben Lengerich, Anna Goldenberg, and Rich Caruana (2021). "How Interpretable and Trustworthy Are GAMs?" In: *Proceedings of the 27th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 95–105.

Chen, Tianqi and Carlos Guestrin (2016). "XGBoost: A Scalable Tree Boosting System". In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.

Dua, Dheeru and Casey Graff (2019). *UCI Machine Learning Repository*. `https://archive.ics.uci.edu/ml`.

Friedman, Jerome H. (2001). "Greedy Function Approximation: A Gradient Boosting Machine". In: *The Annals of Statistics* 29.5, pp. 1189–1232.

Friedman, Jerome H. and Bogdan E. Popescu (2008). "Predictive Learning via Rule Ensembles". In: *The Annals of Applied Statistics* 2.3, pp. 916–954.

Hastie, Trevor and Robert Tibshirani (1986). "Generalized Additive Models". In: *Statistical Science* 1.3, pp. 297–310.

Hooker, Giles (2007). "Generalized Functional ANOVA Diagnostics for High-Dimensional Functions of Dependent Variables". In: *Journal of Computational and Graphical Statistics* 16.3, pp. 709–732.

Hooker, Giles, Lucas Mentch, and Siyu Zhou (2021). "Unrestricted Permutation Forces Extrapolation: Variable Importance Requires at Least One More Model, or There Is No Free Variable Importance". In: *Statistics and Computing* 31.6, p. 82.

Lengerich, Benjamin, Sarah Tan, Chun-Hao Chang, Giles Hooker, and Rich Caruana (2020). "Purifying Interaction Effects with the Functional ANOVA: An Efficient Algorithm for Recovering Identifiable Additive Models". In: *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS)*. Vol. 108. PMLR, pp. 2402–2412.

Lou, Yin, Rich Caruana, and Johannes Gehrke (2012). "Intelligible Models for Classification and Regression". In: *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 150–158.

Lou, Yin, Rich Caruana, Johannes Gehrke, and Giles Hooker (2013). "Accurate Intelligible Models with Pairwise Interactions". In: *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 623–631.

Lundberg, Scott M. and Su-In Lee (2017). "A Unified Approach to Interpreting Model Predictions". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 30.

Molnar, Christoph (2022). *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. 2nd. christophm.github.io.

Nori, Harsha, Samuel Jenkins, Paul Koch, and Rich Caruana (2019). "InterpretML: A Unified Framework for Machine Learning Interpretability". In: *arXiv preprint*. arXiv: 1909.09223.

Sobol', Ilya M. (2001). "Global Sensitivity Indices for Nonlinear Mathematical Models and Their Monte Carlo Estimates". In: *Mathematics and Computers in Simulation* 55.1–3, pp. 271–280.

Strobl, Carolin, Anne-Laure Boulesteix, Thomas Kneib, Thomas Augustin, and Achim Zeileis (2008). "Conditional Variable Importance for Random Forests". In: *BMC Bioinformatics* 9.1, p. 307.

Appendix A. Additional Experimental Details

A.1. XGBoost Hyperparameters

Table A.1 lists the XGBoost hyperparameters used across all experiments. The key constraint is `max_depth=2`, which ensures that each tree uses at most two features and can be exactly decomposed into main effects and pairwise interactions.

Table A.1.: XGBoost hyperparameters used in all experiments.

Parameter	German Credit	Stress-Test
`n_estimators`	120	60
`max_depth`	2	2
`learning_rate`	0.1	0.1
`reg_lambda` (L2)	0	0
`subsample`	1.0	1.0
`colsample_bytree`	1.0	1.0

A.2. Purification Convergence

In all experiments, the purification algorithm converges to the specified tolerance ($\varepsilon = 10^{-10}$) within 5–15 sweeps per interaction matrix. The majority of mass transfer occurs in the first sweep, consistent with the theoretical convergence guarantees described in Section 4.1. No convergence failures were observed across any experimental run.

A.3. Reproducibility

All experiments are fully reproducible. The complete source code, experiment scripts, and result generation pipeline are available at:

`https://github.com/HaykTarkhanyan/fanova_purification`

To reproduce all experiments and generate result files:

```
pip install -r requirements.txt
python run_all_experiments.py
```

The stress-test experiments use 3 random seeds per $(\rho, \text{function})$ combination, and all results are reported as means over these seeds.

Appendix B. Use of AI Tools

In the preparation of this report and the accompanying implementation, AI-assisted tools were used for the following purposes:

- **Code development:** AI coding assistants were used to help write and debug Python code, particularly for the XGBoost tree decomposition and plotting utilities.
- **Writing assistance:** AI tools were used to assist with drafting and editing the text of this report, including structuring arguments, improving clarity, and ensuring consistent notation.
- **Literature search:** AI tools were used to identify relevant prior work and verify bibliographic details.

All AI-generated content was reviewed, verified, and edited by the author. The scientific contributions—the experimental design, the choice of metrics, the interpretation of results, and the critical analysis of ethical concerns—reflect the author's own judgment.

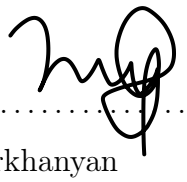
Declaration of Authorship

I hereby declare that this report is my own work and that I have not used any sources or aids except the ones declared or referenced. Text passages quoted verbatim or near-verbatim from external sources, as well as figures taken from or based on external sources, are marked as such.

I further declare that this report has not been submitted to any other examination board in the same or similar form.

Yerevan, Armenia; 01.03.2026

Place, Date



Aik Tarkhanyan